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Statistical Modelling and Inference

Week 2, Lecture 2

- **Outline:**
- **i. Confidence intervals for two population means**
- **ii. Testing of hypotheses for two population means**

Comparison of 2 Population Means

- Goal: To compare 2 populations/treatments with respect to a numerical outcome
- Sampling Design: Independent Samples (Parallel Groups) vs Paired Samples (Crossover Design)
- Data Structure: Normal or Non-normal
- Sample Sizes: Large ($n_1, n_2 > 20$) vs Small

Independent Samples

- Units in the two samples are different
- Sample sizes may or may not be equal
- Large-sample inference based on Normal Distribution
- Small-sample inference depends on distribution of individual outcomes

Parameters/Estimates (Independent Samples)

- Parameter: $\mu_1 - \mu_2$
- Estimator: $\bar{Y}_1 - \bar{Y}_2$
- Estimated standard error: $\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$
- Shape of sampling distribution:
 - Normal if data are normal
 - Approximately normal if $n_1, n_2 > 20$
 - Non-normal otherwise (typically)

Large-Sample Test of $\mu_1 - \mu_2$

- Null hypothesis: The population means differ by Δ_0 (which is typically 0): $H_0 : \mu_1 - \mu_2 = \Delta_0$
- Alternative Hypotheses:
 - 1-Sided: $H_A : \mu_1 - \mu_2 > \Delta_0$
 - 2-Sided: $H_A : \mu_1 - \mu_2 \neq \Delta_0$
- Test Statistic:

$$z_{obs} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Large-Sample Test of $\mu_1 - \mu_2$

- **Decision Rule:**

1-sided alternative $H_A : \mu_1 - \mu_2 > \Delta_0$

- If $z_{\text{obs}} \geq z_{\alpha} \implies$ Conclude $\mu_1 - \mu_2 > \Delta_0$
- If $z_{\text{obs}} < z_{\alpha} \implies$ Do not reject $\mu_1 - \mu_2 = \Delta_0$

2-sided alternative $H_A : \mu_1 - \mu_2 \neq \Delta_0$

- If $z_{\text{obs}} \geq z_{\alpha/2} \implies$ Conclude $\mu_1 - \mu_2 > \Delta_0$
- If $z_{\text{obs}} \leq -z_{\alpha/2} \implies$ Conclude $\mu_1 - \mu_2 < \Delta_0$
- If $-z_{\alpha/2} < z_{\text{obs}} < z_{\alpha/2} \implies$ Do not reject $\mu_1 - \mu_2 = \Delta_0$

Large-Sample Test of $\mu_1 - \mu_2$

- Observed Significance Level (P -Value)

1-sided alternative $H_A : \mu_1 - \mu_2 > \Delta_0$

- $P = P(z_{\text{obs}} > z_{\alpha})$ (From the std. Normal distribution)

2-sided alternative $H_A : \mu_1 - \mu_2 \neq \Delta_0$

- $P = 2P(z_{\text{obs}} > |z_{\alpha/2}|)$ (From the std. Normal distribution)

- If $P\text{-Value} \leq \alpha$, then reject the null hypothesis

Large-Sample $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$

- Confidence Coefficient $(1-\alpha)$ refers to the proportion of times this rule would provide an interval that contains the true parameter value $\mu_1 - \mu_2$ if it were applied over all possible samples

- Rule:

$$(\bar{y}_1 - \bar{y}_2) \pm z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

Example: Vitamin C for Common Cold

- Outcome: Number of Colds During Study Period for Each Student
- Group 1: Given Placebo

$$\bar{y}_1 = 2.2 \quad s_1 = 0.12 \quad n_1 = 155$$

- Group 2: Given Ascorbic Acid (Vitamin C)

$$\bar{y}_2 = 1.9 \quad s_2 = 0.10 \quad n_2 = 208$$

2-Sided Test to Compare Groups

- $H_0: \mu_1 - \mu_2 = 0$ (No difference in trt effects)
- $H_A: \mu_1 - \mu_2 \neq 0$ (Difference in trt effects)
- Test Statistic:
$$z_{obs} = \frac{(2.2 - 1.9) - 0}{\sqrt{\frac{(0.12)^2}{155} + \frac{(0.10)^2}{208}}} = \frac{0.3}{0.0119} = 25.3$$
- Decision Rule ($\alpha=0.05$)
Conclude $\mu_1 - \mu_2 > 0$ since $z_{obs} = 25.3 > z_{.025} = 1.96$

95% Confidence Interval for $\mu_1 - \mu_2$

- Point Estimate: $\bar{y}_1 - \bar{y}_2 = 2.2 - 1.9 = 0.3$
- Estimated Std. Error: $\sqrt{\frac{(0.12)^2}{155} + \frac{(0.10)^2}{208}} = 0.0119$
- Critical Value: $z_{.025} = 1.96$
- 95% CI:

$$(\bar{y}_1 - \bar{y}_2) \pm z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

$$0.3 \pm 1.96(0.0119)$$

$$\Rightarrow 0.30 \pm 0.023$$

$$\Rightarrow (0.277, 0.323)$$

Small-Sample Test for $\mu_1 - \mu_2$ Normal Populations

- **Case 1: Common Variances ($\sigma_1^2 = \sigma_2^2 = \sigma^2$)**
- Null Hypothesis: $H_0 : \mu_1 - \mu_2 = \Delta_0$
- Alternative Hypotheses:
 - 1-Sided: $H_A : \mu_1 - \mu_2 > \Delta_0$
 - 2-Sided: $H_A : \mu_1 - \mu_2 \neq \Delta_0$
- Test Statistic:(where S_p^2 is a “pooled” estimate of σ^2)

$$t_{obs} = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Small-Sample Test for $\mu_1 - \mu_2$

Normal Populations

- Decision Rule: (Based on t -distribution with df $v = n_1 + n_2 - 2$)

1-sided alternative

- If $t_{\text{obs}} \geq t_{\alpha, v} \implies$ Conclude $\mu_1 - \mu_2 > \Delta_0$
- If $t_{\text{obs}} < t_{\alpha, v} \implies$ Do not reject $\mu_1 - \mu_2 = \Delta_0$

2-sided alternative

- If $t_{\text{obs}} \geq t_{\alpha/2, v} \implies$ Conclude $\mu_1 - \mu_2 > \Delta_0$
- If $t_{\text{obs}} \leq -t_{\alpha/2, v} \implies$ Conclude $\mu_1 - \mu_2 < \Delta_0$
- If $-t_{\alpha/2, v} < t_{\text{obs}} < t_{\alpha/2, v} \implies$ Do not reject $\mu_1 - \mu_2 = \Delta_0$

Small-Sample Test for $\mu_1 - \mu_2$ Normal Populations

- Observed Significance Level (P -Value)

1-sided alternative

- $P = P(t > t_{\text{obs}})$ (From the t_v distribution)

2-sided alternative

- $P = 2P(t > |t_{\text{obs}}|)$ (From the t_v distribution)

- If $P\text{-Value} \leq \alpha$, then reject the null hypothesis

Small-Sample $(1-\alpha)100\%$ Confidence Interval for $\mu_1 - \mu_2$ – Normal Populations

- Confidence Coefficient $(1-\alpha)$ refers to the proportion of times this rule would provide an interval that contains the true parameter value $\mu_1 - \mu_2$ if it were applied over all possible samples

- Rule:
$$(\bar{y}_1 - \bar{y}_2) \pm t_{\alpha/2, v} \sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

- Interpretations same as for large-sample CI's

Example - Scalp Wound Closure

- Group1: Stapling ($n_1=15$)
- Group2: Suturing ($n_2=16$)

	Stapling (i=1)	Suturing (i=2)
Mean	96.92	96.31
Std Dev	7.51	8.06
Sample Size	15	16

- a. Conduct a 2-sided test of whether mean scores differ
- b. Construct a 95% Confidence Interval for true difference

Example - Scalp Wound Closure

α : test of whether mean scores differ

$H_0: \mu_1 - \mu_2 = 0$ (mean scores do not differ)

$H_A: \mu_1 - \mu_2 \neq 0$ (mean scores differ)

Test statistics:

$$S_p^2 = \frac{(15-1)(7.51)^2 + (16-1)(8.06)^2}{15+16-2} = 60.83$$

$$TS : t_{obs} = \frac{96.92 - 96.31}{\sqrt{60.83 \left(\frac{1}{15} + \frac{1}{16} \right)}} = \frac{0.61}{2.80} = 0.22$$

$$RR : |t_{obs}| \geq t_{.025, 29} = 2.045 \quad \text{Conclusion: } 0.22 < 2.045$$

$$b: 95\%CI : 0.61 \pm 2.045(2.80) \equiv 0.61 \pm 5.73 \equiv (-5.12, 6.34)$$

Inference Based on Paired Samples (Crossover Designs)

- Setting: Each treatment is applied to each subject or pair (preferably in random order)
- Data: d_i is the difference in scores (Trt₁-Trt₂) for subject (pair) i
- Parameter: μ_d - Population mean difference
- Sample Statistics:

$$\bar{d} = \frac{\sum_{i=1}^n d_i}{n} \quad s_d^2 = \frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n-1} \quad s_d = \sqrt{s_d^2}$$

Test Concerning μ_d

- Null Hypothesis: $H_0: \mu_d = \Delta_0$ (almost equal to 0)
- Alternative Hypotheses:
 - 1-Sided: $H_a: \mu_d > \Delta_0$
 - 2-Sided: $H_a: \mu_d \neq \Delta_0$
- Test Statistic:

$$t_{obs} = \frac{\bar{d}}{s_d / \sqrt{n}}$$

Test Concerning μ_D

Decision Rule: (Based on t -distribution with $v=n-1$ df)

1-sided alternative

If $t_{\text{obs}} \geq t_{\alpha, v} \implies$ Conclude $\mu_d > \Delta_0$

If $t_{\text{obs}} < t_{\alpha, v} \implies$ Do not reject $\mu_d = \Delta_0$

2-sided alternative

If $t_{\text{obs}} \geq t_{\alpha/2, v} \implies$ Conclude $\mu_d > \Delta_0$

If $t_{\text{obs}} \leq -t_{\alpha/2, v} \implies$ Conclude $\mu_d < \Delta_0$

If $-t_{\alpha/2, v} < t_{\text{obs}} < t_{\alpha/2, v} \implies$ Do not reject $\mu_d = \Delta_0$

Confidence Interval: for μ_d

$$\bar{d} \pm t_{\alpha/2, v} \left(\frac{s_d}{\sqrt{n}} \right)$$

Example: Evaluation of students in Mathematics

- Subjects: Students who take Mathematics as course in second semester.
- Data: d_i = difference (in both evaluation tests) for student i
- Summary Statistics:

$$\bar{d} = 1.77 \quad s_d = 1.48 \quad n = 13$$

Example: Evaluation of students in Mathematics

- 2-sided test for differences in both tests,
- use ($\alpha=0.05$)
- $H_0: \mu_d = 0$ $H_a: \mu_d \neq 0$

- **Test statistics:**

$$TS : t_{obs} = \frac{1.77}{\frac{1.48}{\sqrt{13}}} = \frac{1.77}{0.41} = 4.31$$

$$RR : |t_{obs}| \geq t_{.025,12} = 2.179$$

Confidence Interval:

$$95\%CI : 1.77 \pm 2.179(0.41) \equiv 1.77 \pm 0.89 \equiv (0.88, 2.66)$$

Assignment 1

Due Date 27 March 2026 (5:20 pm)

Covers material:

- i. Biased and mean square error
- ii. Confidence intervals for one population mean
- iii. Testing of hypotheses for one population mean
- iv. Confidence intervals for two population means
- v. Testing of hypotheses for two population means

Involves:

Doing all work by handwritten and scanned in as a pdf and please upload to BB

Assignment 1

Due Date 27 March 2026 (5:20 pm)

Q1: Let $Y_1, Y_2, \dots, Y_n$ be a random sample with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2$.

Show that

- i.* $\bar{Y}$ is an unbiased estimator of population mean μ .
- ii.* S^2 is an unbiased estimator of population variance σ^2 .

use,

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Assignment 1

Due Date 27 March 2026 (5:20 pm)

Q2: Researchers are concerned about the impact of students working while they are enrolled in classes. A survey of 200 students provides a sample mean of 7.10 hours a week worked, with the standard deviation is about 3 hours.

- i. What is a 95% confidence interval for population mean?
- ii. At 5% level of significance test the hypotheses that students worked more than 7 hours a week.

(Find the value of the test statistic, find the rejection region for the test, Conduct the test and state your conclusions.)

Assignment 1

Due Date 27 March 2026 (5:20 pm)

Q3: Independent random samples of $n_1 = 29$ and $n_2 = 18$ observations were selected from two normal populations with unequal variances and found $\bar{Y}_1 = 37.6, \bar{Y}_2 = 35.9, S_1 = 5.2$ and $S_2 = 6.7$.

Suppose you wish to detect a difference between the population means.

- i. State the null and alternative hypotheses for the test.
- ii. Find the rejection region for the test use $\alpha = 0.01$.
- iii. Find the value of the test statistic.
- iv. Calculate the approximate p-value for the test.
- v. Conduct the test and state your conclusions.

**Thank
You**

谢谢

